

European Commission's investigations on cloud computing services under the Digital Markets Act

November 2025 (ongoing case)

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Executive Summary

The report focuses on the implications that the EU Commission gatekeeper designation would have on the gatekeepers. More precisely, it looks at the way the concerns would be addressed, if they would be effectively addressed or if they would create other issues, considering the inherent ever evolving nature of the market and eventually suggests remedies on top of the DMA obligations.

The Commission's three investigations examine the market of cloud computing services within the EU. They test whether Amazon and Microsoft are gatekeepers of their cloud computing services (AWS and Azure, respectively), and if potential anti-competitive measures can be tackled by the DMA (EC, 2025). This report investigates the market's context, key products, market power, and potential barriers to entry.

This report assesses whether AWS and Azure qualify as gatekeepers under the DMA by first defining the relevant market and evaluating its level of contestability. It then identifies how switching costs, network effects, bundling practices and interoperability barriers shape competitive dynamics, and uses this analysis to propose targeted remedies. The analysis draws on the findings of the UK CMA's recently concluded cloud services market investigation, using it as a benchmark to inform and support the assessment.

SECTION I : Market Description

Context: What is Cloud Computing?

Cloud computing is the outsourcing of physical IT infrastructure at scale. Instead of maintaining their own servers and storage hardware, businesses rent computing capacity from third-party providers operating large data centres on demand. This shift has made cloud infrastructure critical in the modern digital economy. The Commission recognises that these *"cloud services must be provided in a fair, open and competitive environment"* (EC, 2025), to foster innovation and improve Europe's strategic autonomy.

It is identified as a key market as the cloud is *"crucial for AI development."* Training LLMs requires data centers and cloud software that only hyperscale providers can provide at the necessary scale. The race for Generative AI has only accelerated this demand for cloud (van der Vlist, 2024), creating a concentrated market that the Commission's investigations are designed to address (EC, 2025).

Gatekeeper Designation & Definition

Article 3 of the Digital Markets Act defines the criteria for "gatekeeper" designation. This status is the prerequisite for regulatory action because it determines whether the DMA's obligations under Articles 5 and 6 apply to AWS and Azure. Without this designation, the European Commission lacks the legal power to address the bundling, switching costs, network effects and barriers to entry identified in this report.

The DMA establishes three quantitative thresholds for automatic designation. These include an EEA turnover of at least €7.5 billion, 45 million monthly active end users in the EU, and 10,000 yearly active business users (DMA, Article 3). The Commission confirmed in 2025 that AWS and Azure do not meet all of these thresholds (EC, 2025). However, the Commission retains the authority to designate a firm qualitatively, if either firm has *"a significant impact on the internal market, ... enjoys an entrenched and durable position,"* and firms remain an *"important gateway for business users to reach end users"* (DMA, Article 3) The Commission has successfully used this qualitative designation before for Apple's iPadOS.

This report focuses on the first two investigations to determine if these competitive concerns exist in the cloud market. The third investigation, which evaluates the ultimate effectiveness of DMA enforcement powers, remains secondary until the threshold for designation is proven.

Market Size & Growth

Global public cloud end-user spending reached \$595.65 billion in 2024, forecast to grow to \$723.42 billion in 2025 (Gartner, 2024), representing a six-fold increase over the past decade. In Europe, the total cloud infrastructure market is projected to reach \$430 billion by the end of the decade (Statista, 2025), driven by US providers investing an estimated €10 billion every quarter in European cloud infrastructure (Synergy Research, 2024). Whilst this level of capital has driven market growth, the scale raises concerns about contestability, as the endogenous sunk costs required to compete credibly act as a structural barrier that potential entrants may struggle to overcome.

The Products: IaaS, PaaS and SaaS

To understand the market dynamics, it is essential to understand the products that AWS and Azure provide. Cloud services are delivered in distinct layers, each building upon the one below. The CMA's parallel investigation (CMA, 2025) identified IaaS and PaaS as the key cloud products for assessing contestability.

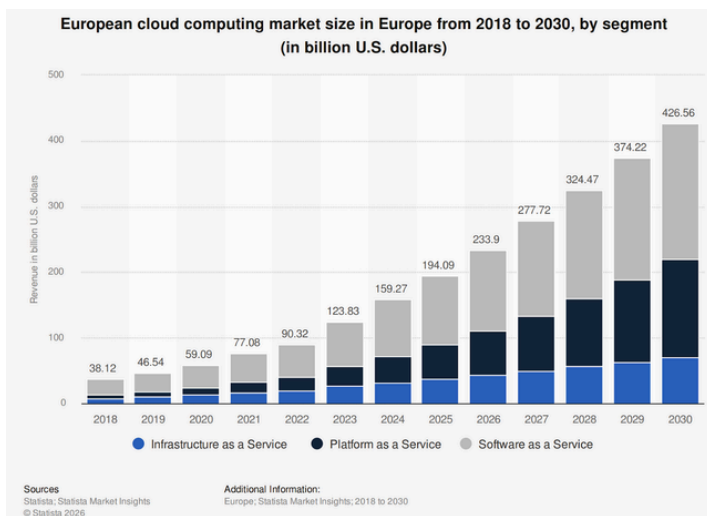


Figure 1: European Cloud Computing Market Size Projections by Segment

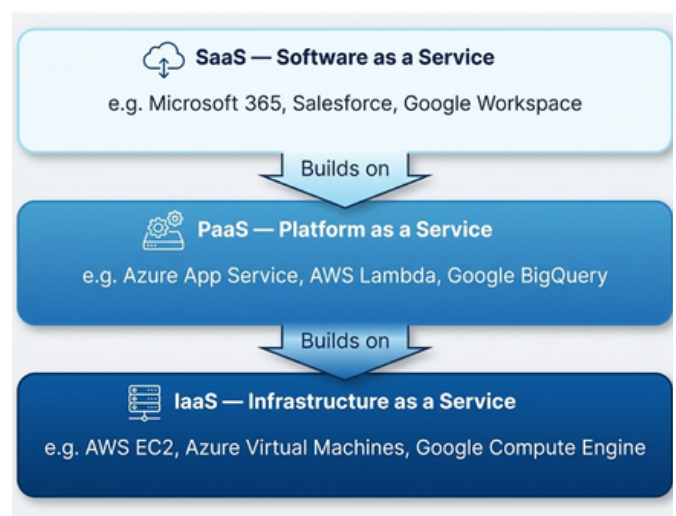


Figure 2: Cloud Computing Products. IaaS is the foundation, PaaS and SaaS are dependent on IaaS.

IaaS is the foundational layer, referring to the provision of raw computing services, such as services, storage and networking, rented by customers over the internet. AWS's Elastic Compute Cloud and Microsoft's Azure Virtual Machines are the flagship IaaS products.

PaaS operates a layer above, providing the services that can be used to develop, test, run and manage applications in the cloud (CMA, 2025), with AWS Lambda and Azure App Service the leading examples. SaaS provision of finished application software sits atop this.

PaaS and SaaS providers are entirely dependent on IaaS Infrastructure. This asymmetric dependence is an issue, as while an IaaS provider can readily offer PaaS and SaaS services on top of existing infrastructure, the opposite is difficult due to the huge physical capital requirements involved, leading to an advantage for IaaS providers. On SaaS Microsoft has M365, while Amazon primarily competes in PaaS and IaaS with AWS Lambda, Cognito and Connect.

Market Structure: A Highly Concentrated Oligopoly

The underlying European cloud infrastructure (IaaS) market is a concentrated oligopoly with persistent market shares. Motta associates this with durable dominance rather than competitive equilibrium (Motta, 2004). Globally, the three US hyperscalers collectively hold approximately 65% of worldwide cloud infrastructure (Statista, 2026) and the European market should mirror this closely.

Provider	Global IaaS Share (Gartner, 2024)	UK IaaS Share (CMA, 2025)	European IaaS Share (2024)
Amazon Web Services (AWS)	28%	30–40%	30%
Microsoft Azure	21%	30–40%	25%
Google Cloud	14%	10–15%	10%
US Hyperscalers	63%	70%	65%

Figure 3: IaaS Market Shares by Provider. **Sources:** Statista (2026); CMA (2025); European Commission (2025)

Lacking specific EU firm-level data, CMA (2025) figures provide the clearest proxy. AWS holds **30–35%** and Azure **25–30%**. The persistence of these market shares amid market growth signals structural advantages. In a contestable market, entry and price competition would erode such dominance. The decline of European IaaS providers despite a six-fold market expansion since 2017, as shown by the combined share of firms like SAP, Deutsche Telekom, and OVH collapsing from **29% to 15%** in 2024 (EC, 2025). While their absolute revenue has grown, their relative marginalization perfectly illustrates Motta's (2004) thesis: persistent market share in a growing sector is the definitive indicator of entrenched dominance and a lack of contestability.

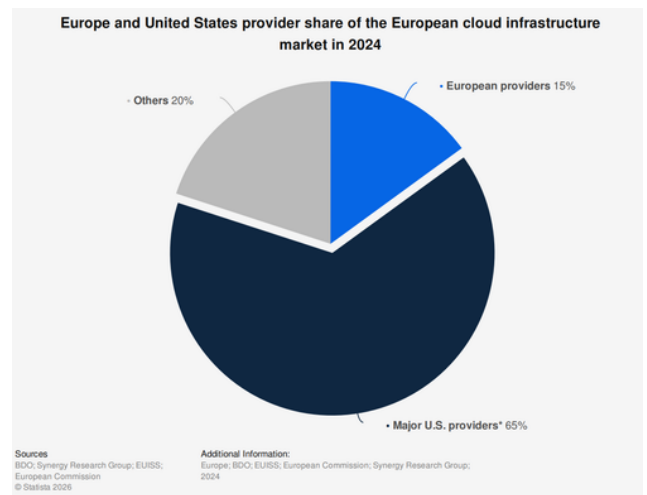


Figure 4: European vs US Cloud Provider Market Share

Barriers to Entry

The cloud market combines structural barriers to entry with firm practices that may further reduce contestability. Structural features such as high fixed costs, sunk investment, and economies of scale already favour large incumbents. In addition, pricing, contractual, and technical practices including egress fees, bundling, and interoperability frictions may further raise switching costs and make rival entry or expansion more difficult.

- Egress Fees and Switching Costs:** Hyperscalers charge fees of up to €0.08/GB for transferring data out of their infrastructure (CMA, 2025), creating a financial penalty for switching. Once an enterprise's workloads are embedded in a provider's infrastructure, the cost of exit exceeds any price a rival could offer, enabling incumbents to sustain prices above competitive levels (Klemperer, 1995).
- Tying and Bundling:** Hyperscalers discount compute services only when purchased alongside storage, identity management and networking products (all IaaS components). This bundle extends upwards into PaaS tools like managed databases, AI services, analytics platforms, and even further into SaaS, making disaggregation economically irrational for customers. This hurts rivals such as Zoom or Heroku, who offer competitive standalone PaaS/SaaS services but cannot replicate the bundled price that combines all three products (EC, 2025).
- API & Incompatibility Barriers:** AWS and Azure provide platforms with their own Application Programming Interfaces (APIs or PaaS) that are incompatible with rival clouds (rival IaaS). Migrating an application built on these services can cost millions, making switching more expensive even where a rival is cheaper (EC, 2025). The Commission is investigating whether this is deliberate, raising serious concerns about entrenched dominance (EC, 2025).

SECTION II : Market Definition

Market definition establishes the boundaries within which the Commission assesses whether AWS and Azure operate as important gateways. Article 2 of the Digital Markets Act (DMA) identifies cloud computing as a Core Platform Service, a distinct category significant enough to warrant regulation (DMA, Article 2: Definitions), but it does not draw the market's boundaries, so this section below establishes those lines.

Demand Side Substitutability

The SSNIP Test and On Premise Limitations: Enterprises are unlikely to substitute away from AWS or Azure in response to a 5 to 10% price increase. While on premise IT is theoretically a substitute, it is not a credible alternative for most modern enterprises. High switching costs and technical lock in create significant demand inelasticity. The elastic scaling, integrated services, and global reach of hyperscalers cannot be replicated on premise at a comparable scale or cost.

Product Differentiation: IaaS SaaS and PaaS: Pure SaaS providers such as Salesforce or Adobe are not substitutes for IaaS providers, because they sell finished software applications rather than underlying compute, storage, and networking capacity. A customer facing a price increase in IaaS would not switch to a software platform, since these products serve entirely different functions.

Providers offering purely PaaS, such as Heroku or Vercel, are also not credible substitutes. APIs mean that migrating from one provider's PaaS to another often requires a simultaneous IaaS migration. This makes IaaS and PaaS inseparable from a customer's perspective. A price rise at the IaaS layer impacts the entire stack, which means both layers lie within the same market only if they are bundled together.

Supply Side Substitutability

Endogenous Sunk Costs and Barriers to Entry: Unlike typical markets where high profits attract entry, entry into IaaS is blocked by massive endogenous sunk costs. Replicating the physical infrastructure of an incumbent requires years of construction and billions in capital expenditure. Consequently, market contestability is unlikely to discipline prices.

Asymmetric Entry Dynamics: The direction of entry is fundamentally asymmetric. An IaaS provider can readily move upward into PaaS and SaaS by building managed services and software applications on top of existing infrastructure. Both AWS and Azure have demonstrated this trajectory. Conversely, a PaaS or SaaS provider cannot move downward into IaaS without first crossing an enormous capital threshold.

Microsoft's transition from a PaaS focused Azure to a full stack provider illustrates that downward movement is not theoretically impossible, but it requires a balance sheet and global infrastructure base unavailable to typical competitors. Therefore, the large population of PaaS providers cannot be counted as potential entrants capable of constraining IaaS pricing. They compete only at higher layers of the stack, leaving the underlying infrastructure market without effective competitive discipline from below.

The Relevant Product Market

Given the inelastic demand and the failure of the SSNIP test across potential substitutes, the relevant product market must be defined strictly. On premise infrastructure lacks the elastic scaling and operational expenditure model required to be a credible substitute. SaaS serves entirely distinct functional needs. Furthermore, the asymmetric barriers to entry and the technical entanglement of IaaS and PaaS necessitate grouping them together.

Therefore, the relevant market is defined exclusively as the provision of bundled public cloud infrastructure services anchored by IaaS. Providers offering solely Software as a Service or solely Platform as a Service fall

entirely outside this boundary. To be considered a participant in this specific market, a provider must supply foundational IaaS. Pure play PaaS, pure play SaaS, and traditional on premise IT are explicitly excluded from this definition.

Geographic Market

While cloud services are delivered globally, the relevant geographic market is the European Economic Area (EEA). The GDPR requires that personal data relating to EU residents be stored and processed within the EEA, meaning enterprises cannot legally substitute EEA-located cloud with infrastructure hosted elsewhere for most regulated work (GDPR, 2026). The GDPR therefore acts as a trade barrier creating a distinct regional market, ensuring the Commission's regulatory perimeter maps onto economic reality.

SECTION III: Theories of Harm

The dominance of AWS and Azure generates tangible harm across three distinct groups: business users and alternative providers; public sector entities exposed to sovereign risk; and end consumers bearing the downstream effects of reduced competition. Each harm channel operates through a distinct mechanism.

1. Business Users, SMEs, and Alternative Cloud Providers

Business users are the most directly affected. Vendor lock-in means that once an enterprise has embedded its operations into a proprietary ecosystem, switching becomes prohibitively costly. Egress fees of up to €0.08/GB (CMA, 2025) impose a direct financial penalty on switching, and proprietary APIs incompatible with rival clouds require costly application rewrites upon migration (EC, 2025). Klemperer (1995) formalises this dynamic: switching costs enable incumbents to exploit locked-in users rather than compete for marginal customers, sustaining prices above competitive levels. Farrell and Klemperer (2007) extend this further, demonstrating that even vigorous ex ante competition does not discipline ex post pricing once customers are locked in, directly applicable to cloud markets where multi-year committed spend agreements entrench provider dependency. Chen, Doraszelski and Harrington (2009) further suggest that incompatibility may be a deliberate strategic choice rather than an incidental feature, raising concerns that interoperability barriers are instruments of foreclosure. Compounding this, the overlapping digital regulatory landscape, spanning GDPR, the AI Act, the Data Act, and NIS2, creates compliance overheads that burden smaller businesses disproportionately, functioning as a structural barrier that reinforces incumbent advantage without any deliberate anticompetitive actions (European Parliament, 2025).

2. Public Sector Entities and Geopolitical Risk

European public sector entities face a harm that is simultaneously legal, operational, and geopolitical. The U.S. CLOUD Act can require U.S.-headquartered providers to disclose data within their possession, custody, or control, regardless of where that data is stored, creating a conflict with GDPR's requirement that personal data not be transferred to third countries without adequate safeguards (GDPR, Article 46). The European Data Protection Board (2020) has confirmed that contractual arrangements alone cannot resolve this conflict, and Linklaters (2019) concludes that the CLOUD Act and GDPR are structurally incompatible, no contractual or technical measure can simultaneously satisfy both regimes. Critically, this vulnerability has been demonstrated at the highest institutional level: the EDPS found that the European Commission itself breached GDPR through its own use of Microsoft 365, identifying failures including lack of control over personal data, unclear transfers outside the EEA, and insufficient safeguards against third-country access (EDPS, 2024). That the EU's own institutions cannot use Microsoft 365 in full GDPR compliance powerfully illustrates the structural irreconcilability of US cloud dependency with European data protection law. Bradford (2020) characterises this as the "Brussels Effect in reverse": US extraterritorial legislation undermining EU regulatory sovereignty from within. The consequences have further materialised in practice: US sanctions previously blocked ICC personnel from accessing Microsoft cloud services, demonstrating that infrastructure dependence can translate directly into operational disruption for bodies exercising sovereign functions (Michels, 2026).

3. End Consumers

Whilst end consumers are indirect victims, their harm is neither trivial nor speculative. Weyl and Fabinger (2013) demonstrate that pass-through rates are typically high in markets with concentrated structure and inelastic demand, both features of cloud-dependent sectors, meaning higher infrastructure costs borne by businesses are passed through significantly to consumers across financial services, healthcare, retail, and media. Beyond price, Aghion et al. (2005) establish empirically that markets which are too concentrated generate insufficient innovation incentives, and the persistent foreclosure of potential entrants reduces the innovative dynamism that would otherwise generate new services and lower prices over time. As Crémer, de Montjoye and Schweitzer (2019) warn, these harms are diffuse and indirect, making them systematically difficult to quantify but no less real and as digital services become increasingly embedded in everyday life, their significance will only grow.

SECTION IV: Competition Concerns

Analysis of Core Competition Issues in Cloud Computing

The CMA report is used as a methodological benchmark as it provides a complete, evidence-based assessment of the cloud market under the Enterprise Act. The EU investigation is ongoing and focuses on assessing whether similar market features and practices exist that may limit competitiveness and fairness (European Commission, 2025).

The CMA Final Decision Report shows that the CMA gathered evidence by consulting a large number of stakeholders, including UK customers, major cloud providers, independent software vendors (ISVs), and academics. The CMA focused primarily on qualitative evidence, expressing concerns that quantitative surveys in this complex market lacked validity. In addition, the CMA conducted rigorous financial assessments, including a profitability analysis comparing Return on Capital Employed (ROCE) to Weighted Average Cost of Capital (WACC). To investigate the licensing tie, the CMA analysed quantitative pricing data by comparing the effective prices Microsoft charges its own Azure customers for Windows Server and SQL Server against the higher input prices it charges users of rival cloud providers for the same software (CMA, 2025).

1. Tying and Bundling through Software Licensing

The CMA identifies that the dominant cloud providers can utilise their owned and operated services to increase market power, and its detailed findings on tying relate specifically to Microsoft's software licensing practices. The CMA concludes that Microsoft has significant market power in several software products used as key inputs to cloud services, and that customers may face significant differences in price and/or quality depending on whether they use those Microsoft products on Azure versus on AWS or Google. Most importantly for foreclosure theory, the CMA finds that for some Microsoft software products, the licensing fee AWS and Google must pay Microsoft in order to offer those products on their own clouds can exceed the price Microsoft itself charges some customers for using the same software on Azure.

To interpret the mechanics of this market leverage, we draw on the framework of de Cornière and Taylor (2021), which only partially fits the present case but is useful for illustrating how control over an upstream input can shape competition in a downstream market. Here, Microsoft supplies a quasi-essential software input, such as Windows Server and SQL Server (Component A), while also competing downstream in cloud infrastructure (Component B) through Azure against rival providers such as AWS and Google. AWS and Google must license this software on their IaaS through agreements such as the Services Provider Licensing Agreement (SPLA). The CMA finds that the prices charged under the SPLA are significantly higher than the effective prices Microsoft offers to its own Azure customers who benefit from discounts such as the Azure Hybrid Benefits. Business customers act as downstream firms that serve end-users and face integration constraints, as in practice they tend to standardise around one primary software environment within their systems. The market exhibits retail complementarity: if the core infrastructure is used, downstream demand is high (Q), and profits are Π ; without it, demand falls to $q < Q$ and profits to $\pi < \Pi$.

Without tying, a rival infrastructure provider (B2) expects to be chosen together with the essential software input (A), allowing it to compete for the full set of Microsoft-dependent workloads.

$$F_{B_2} = -r_2 Q$$

If the dominant firm makes the combination of A and B1 more attractive than A and B2, demand shifts away from the rival infrastructure provider. In the model, this is represented by a fall in the rival's expected market size from Q to q , which reduces its incentive to compete.

$$F_{B_2} = -r_2 q$$

The downstream firm adopts the bundle if:

$$\Pi - F_1 \geq \pi + r_2 q$$

Which implies:

$$F_1 = \Pi - \pi + r_2 q$$

The dominant firm's profit is:

$$V_1 = (\Pi - \pi - r_2 q) + r_1 Q$$

Bundling is profitable if:

$$r_1 Q > r_2 q$$

In the de Cornière and Taylor (2021) model, the rival faces a reduced expected market size and therefore offers a lower slotting fee. Microsoft holds significant market power over essential software, including Windows Server and SQL Server, which function as critical inputs that cannot be substituted away from.

Choi, Jeon, and Whinston (2025) show that when a monopolist cannot fully extract consumer surplus in one market, tying can create a "quasi-installed base" advantage in the tied market, potentially excluding rivals but with ambiguous welfare effects. By tying infrastructure to essential software, this strategy may limit competition and create barriers to entry.

2. Network Effects and Ecosystem Reinforcement

Although the CMA's core findings focus on concentration, switching barriers, and licensing practices, network effects and installed-base advantages help explain why those concerns may be persistent. At the IaaS layer, the cloud market exhibits strong installed-base advantages. A larger customer base makes a provider's infrastructure more attractive in ways that go beyond price: more enterprises on AWS or Azure mean a deeper pool of engineers trained on provider-specific tools, more third-party integrations built around those architectures, and a stronger base of compliance, certification, and operational familiarity.

Each additional enterprise customer, therefore, strengthens the rationale for the next enterprise to choose the same provider. This creates a self-reinforcing ecosystem dynamic that favours incumbents. At the platform level, a second compounding dynamic emerges with two-sided characteristics. Because enterprise customers are already concentrated on the leading hyperscalers, independent software vendors and tool-builders such as Databricks and Snowflake may have strong incentives to build their products around those platforms to access that customer base. This is consistent with a richer ecosystem of integrated tools on leading platforms than on smaller rivals, which in turn attracts more enterprise customers, reinforcing the original advantage.

The key harm channel is reduced contestability: when ecosystems reinforce incumbency, rivals must overcome not only price/quality gaps but also coordination advantages embedded in training, integrations and complementary service availability. This contributes to reducing contestability and sustained market concentration, particularly when combined with switching barriers identified in the CMA investigation.

3. Switching Costs and Interoperability Obstacles

Ecosystem dominance is further protected by switching costs, which the CMA identifies as a major barrier to competitive switching and multi-cloud adoption. These switching costs are both financial and technical. Financial switching costs have historically included egress fees charged for transferring data out of a provider's infrastructure, while technical switching costs arise when customers build applications around provider-specific

APIs, managed services, and internal skill sets. Under the EU Data Act, switching charges such as egress fees are being phased out, while providers are also required to facilitate switching and interoperability (European Commission, 2026); technical migration frictions, however, are reduced rather than eliminated outright.

Switching-cost theory provides the demand-side foundation. Klemperer (1995) emphasizes that switching costs shape dynamic price competition because firms can exploit locked-in users rather than compete aggressively for marginal customers. An enterprise that has built its operations around AWS or Azure may have written applications against provider-specific APIs, embedded workloads in proprietary managed services, and trained internal teams around that environment. Migrating to another provider may therefore require code adaptation, workforce retraining, and extensive retesting of applications and security processes. Even where a rival offers a lower nominal price, the total cost of migration may render that offer commercially unattractive.

The most policy-relevant extension is that compatibility itself may be strategic. Chen, Doraszelski, and Harrington (2009) model markets with network effects in which firms may choose whether to remain compatible with rivals; a firm with a large installed base may have weaker incentives to maintain compatibility where incompatibility helps preserve its advantage. Applied to cloud markets, this suggests that interoperability cannot be assumed to emerge naturally through competition alone.

Critical Evaluation

According to Choi, Jeon, and Whinston (2021), a dominant firm can bypass the traditional Chicago School argument that “bundling is harmless” by leveraging the unexploited consumer surplus from its monopoly product (legacy software) to automatically generate a quasi-installed base in a related, competitive market. By making it financially punishing to use rival clouds, Microsoft forces customers into Azure without needing to compete on the standalone merits of its infrastructure. Further, de Cornière and Taylor (2021) demonstrate that upstream bundling destroys rivals’ incentives and willingness to compete for essential downstream channels. In the cloud market, this means competitors are structurally disadvantaged in bidding for enterprise workloads because their input costs are inflated, which marginalises their competitive offerings. The market does not fully reflect the model, as AWS and Google remain competitive due to differentiation and multi-cloud adoption, implying only partial foreclosure. However, an evaluation of the economic literature suggests that the European Commission’s concerns may be overstated if regulators mistakenly assume that a market with many cloud providers is inherently efficient. The economic model of Choi (2004) explicitly demonstrates that when R&D and infrastructure require massive fixed costs, tying and market concentration can enhance overall social welfare by eliminating inefficient, duplicative research efforts among smaller rivals. The UK CMA’s findings support this efficiency rationale, acknowledging that volume-based bundles and Committed Spend Agreements (CSAs) are not inherently harmful; rather, they provide hyperscalers with the predictable demand necessary to securely fund multi-billion-pound infrastructure investments. Notably, the evidence underpinning the foreclosure mechanism is concentrated on Microsoft’s software licensing practices, with no equivalently detailed findings presented for AWS or Google in the CMA analysis.

Furthermore, the methodological choices of the investigation must be scrutinised. The CMA openly prioritised qualitative evidence because cloud procurement is complex, but that methodological choice increases the burden to triangulate stakeholder claims with measurable indicators. For the EU investigations, a strong improvement would be to exploit natural experiments: licensing-policy changes, time-limited free switching programmes, or service deprecations that create observable shocks to switching and multi-cloud behavior.

Lastly, regulatory timing is a major challenge. The DMA adopts a preventive, ex ante logic rather than requiring a full demonstration of harm in the ex post abuse-of-dominance sense. That may be justified in a market where installed-base advantages, ecosystem effects, and switching costs are cumulative. However, the same preventive logic also creates a risk of intervention that is too rigid for a market evolving rapidly. Multi-cloud adoption is expanding, and tools such as Kubernetes may reduce some forms of workload-level lock-in by improving portability across environments, even if they do not eliminate cloud dependence (CNCF, 2017; CNCF, 2023). At the same

time, AI-related demand is reshaping competitive dynamics and increasing the strategic importance of cloud infrastructure. The challenge for regulators is therefore whether intervention is warranted and if it can remain well targeted in a market whose sources of competitive advantage may shift faster than the regulatory response.

SECTION V: Remedies

Remedies are designed to correct the negative impact of anti-competitive behaviour in markets. They play a central role in enforcing competition law, and act as a benchmark for measuring the efficiency of the overall system (Gerard and Komninos, 2020). According to the European Commission, remedies should eliminate competition concerns identified by the authority while remaining proportionate, effective, and capable of being implemented in practice (European Commission, 2008).

In digital markets, remedies are particularly challenging to design. The presence of strong network effects, switching costs, and rapid technological change means that poorly designed interventions could fail to restore competition, or unintentionally destroy efficiency gains that benefit consumers (Scott Morton, 2025). In the cloud computing market, these challenges are especially important. As mentioned before, AWS and Azure's dominance is not simply the result of anticompetitive conduct, but also due to the structural features such as large sunk infrastructure costs and proprietary APIs that any effective remedy must engage with directly. Remedies in this context must therefore aim to improve interoperability, reduce customer lock-in, lower barriers to entry for rival providers, and address specific practices such as licensing and bundling that reinforce incumbent positions beyond what market structure alone would produce.

The European Commission's investigations into cloud computing services under the DMA were launched in November 2025 and remain ongoing. While no official remedies have been imposed yet, the theory of similar cases and the remedies imposed in similar cases in the past allows us to evaluate which remedies are likely to be effective in this case.

Types of Remedies

In EU competition policy, remedies are typically classified as structural or behavioural. Structural remedies alter the market structure, while behavioural remedies constrain firm conduct. In digital markets, a subset of behavioural remedies focuses specifically on improving contestability through interoperability, data portability, and switching, which are particularly relevant under the DMA framework.

Structural Remedies

Structural remedies involve changes to the structure or ownership of a firm. They are often "one-off" measures that aim to maintain or restore market structure by creating or enhancing a new competitive player (Wilson, 2020). These interventions can include the divestiture of assets, sale of business units, or separation of vertically integration activities (Walle, 2021, p.26). The European Commission generally prefers structural remedies to behavioural ones, viewing them as more durable and less dependent on monitoring (Ritz, Wünschmann and von Schreitter, 2025). This preference reflects an empirical record based on an analysis of 20 years of antitrust enforcement, which found that structural remedies were always implemented and generally effective, while behavioural remedies tended to be poorly enforced, partially implemented, or simply ineffective (Ritz, Wünschmann and von Schreitter, 2025).

However, structural remedies may be more difficult to implement in complex digital markets (Dorell and Chapman, no year). Digital platforms often operate through integrated ecosystems of services, and breaking up these structures may be technically difficult and potentially disruptive to innovation and investment. Choi argues that in cases involving economies of scale or large fixed costs, concentration of activity in fewer firms may be more efficient as it avoids the duplication in investment (Choi, 2004). A structural remedy that would involve breaking

up AWS or Microsoft Azure risks destroying genuine efficiency gains without necessarily restoring contestability due to the structural barriers to entry which would remain even after divestiture. Furthermore, the DMA’s framework is not primarily designed around structural intervention. Therefore, we focus our analysis on behavioural and market-based remedies.

Behavioural Remedies

Behavioural remedies impose constraints on how a firm conducts itself without changing its structure. Although the commission has favoured structural remedies historically, behavioural remedies are likely to be more appropriate in this case (Walle, 2021, p.10). In cloud computing, the core concerns identified in this report (tying, bundling and switching costs) have a significant behavioural component, making this the more relevant category in our case. Typically, the behavioural remedies imposed by the DMA on digital gatekeepers include adhering to data-sharing requirements and ensuring non-discriminatory access to platforms (Dorell and Chapman, N.A.). These obligations are set to even the playing field and encourage competition in digital markets.

Market-Based Remedies

Market-based remedies aim to improve competitive conditions by reducing barriers to entry and facilitating switching, rather than directly restructuring firms or imposing strict conduct rules. Though not always formally classified alongside structural and behavioural remedies, many DMA obligations fall within this category, including data portability requirements, transparency obligations on pricing and service terms, and interoperability mandates. In our case, these measures could be reinforced by complementary legislation such as the EU Data Act, which further reduces switching costs through interoperability standards and the phased elimination of data egress fees, which would strengthen contestability in cloud markets.

CMA Cloud Services Market Investigation (2025)

The CMA’s cloud services market investigation, concluded in July 2025, is the most directly comparable precedent for the EC’s ongoing investigations. Similarly to our findings, the CMA highlighted high switching costs, where customers are locked into their initial cloud provider by features such as egress fees, technical incompatibilities, and committed spend discounts (CMA, 2025). As a result, the CMA chose not to impose direct remedies, finding that those would carry risks to the effectiveness of the market. Instead, they concluded that a further investigation under the DMCC act was necessary to designate Microsoft and AWS with strategic market status (roughly the equivalent of being designated a gatekeeper under the DMA). However, without imposing them, they do suggest that effective remedies could include:

AEC	Suggested Remedy	Desired Effect
Switching Costs	Price control in the form of a full ban on egress fees for switching or an at-cost cap	Reduce the financial barrier to switching cloud provide
Switching Costs	Mandated API interoperability and portability requirements	Prevent technical lock-in recreating the barrier once financial costs are removed
Bundling and Tying	FRAND licensing obligation requiring Microsoft to price software (e.g. Windows Server, SQL Server) on fair, reasonable and non-discriminatory terms regardless of which cloud it runs on	Eliminate the cost asymmetry between Azure and rival clouds for enterprise customers

Figure 4: Proposed Remedies and Expected Effects Source: CMA, 2025

Effectiveness of Remedies and Final Decision

Evaluating remedies requires assessing whether they address the source of the problem, preserve efficiencies where possible, and can be implemented and enforced effectively in practice. In cloud computing, these criteria are complicated by the mutual reinforcement between switching costs, network effects, and bundling identified in this report. Partial remedies risk being not only incomplete but counterproductive, as firms could shift exclusionary behaviour from the targeted mechanism to an untargeted one.

FRAND-based remedies have been widely used across EU competition regulation to improve access to address power imbalances, although their effectiveness is context-dependent and difficult to operationalise in practice (Mantzari, 2024). The CMA case considered price controls on egress fees and a FRAND-based licensing package as possible remedies, while also stressing that any intervention would need flexibility, monitoring, and the ability to adapt over time (CMA, 2025). This is important because as Chen, Doraszelski and Harrington (2009) show, when firms can choose whether to make their products compatible, they can behave strategically to maintain their advantage from having more users. In a tipping equilibrium, once a firm gains an advantage, rivals have limited incentives to compete aggressively, and dominance is self-reinforced through incompatibility and network effects rather than price alone.

Applied to our case, reducing financial switching costs without simultaneously addressing technical interoperability risks leaves this tipping dynamic intact. For example, if AWS were prevented from charging egress fees, it could respond by increasing storage or compute prices, thereby preserving switching costs through alternative channels. Similarly, a FRAND licensing obligation may reduce cost asymmetries created by Microsoft's licensing practices, but on its own it cannot prevent lock-in from being recreated through proprietary APIs or other technical restrictions. This is consistent with de Cornière and Taylor (2021), who show that bundling strategies can soften rivals' competitive constraints by reducing their incentives to compete for access to users, thereby sustaining market power even in the presence of rivals. Interoperability remedies must therefore complement pricing remedies, rather than follow them.

Structural remedies would, in theory, address more directly the durable market power generated by network effects, but they are not the best option here. The CMA investigation concluded that structural remedies and operational separation would not be effective or proportionate in this market (CMA, 2025, p.636). This conclusion makes sense for our case, where, as mentioned earlier, cloud infrastructure depends on very high fixed costs and strong economies of scale, so forced separation risks eliminating efficiencies without removing the barriers to entry of the market (Choi, 2004).

This report therefore recommends a coordinated set of behavioural and market-based remedies, implemented simultaneously rather than sequentially. The most appropriate remedies would include:

- 1.Regulation or prohibition of egress fees
- 2.Mandatory interoperability requirements
- 3.FRAND licensing obligations for software used across rival clouds

While these remedies apply broadly to cloud providers, their relevance may differ across firms. In particular, FRAND-based licensing obligations are most directly targeted at Microsoft, given its control over essential software inputs such as Windows Server and SQL Server. By contrast, remedies aimed at reducing switching costs and improving interoperability are more uniformly applicable across both AWS and Azure, as both providers rely on proprietary architectures that contribute to customer lock-in. This suggests that, although a common regulatory framework can be applied, the intensity and focus of remedies may need to vary across providers.

Each of these remedies involves trade-offs. FRAND licensing may be difficult to monitor and enforce in practice, particularly where pricing structures are complex. Interoperability obligations can reduce switching costs but may limit incentives for product differentiation. Similarly, banning egress fees may lead providers to adjust pricing

elsewhere, reducing the overall effectiveness of the intervention. These challenges reinforce the need for coordinated and adaptable remedies.

Combined, these remedies target the three mechanisms identified in our analysis (network effects, switching costs, and bundling) while preserving the efficiencies associated with the scale. Since AWS and Azure currently fall outside of the quantitative thresholds of article 3 due to the B2B nature of cloud services, the EC should prioritise the gatekeeper designation via the qualitative route under Article 3(9) as a necessary first step before the full Articles 5 and 6 toolkit can be applied.

SECTION VI: Conclusion

This report has assessed whether AWS and Azure qualify as gatekeepers of cloud computing services under the Digital Markets Act, and whether the competitive concerns identified in the European cloud market warrant regulatory intervention.

The European cloud infrastructure market is characterised by a highly concentrated oligopoly in which three US hyperscalers collectively account for approximately 70% of regional revenues. This concentration is structural rather than temporary. European providers' share collapsed from 29% to 15% between 2017 and 2024, even as the overall market grew sixfold, consistent with Motta's (2004) thesis that persistent market share amid growth is the definitive indicator of entrenched dominance rather than competitive equilibrium.

Three interrelated mechanisms drive this outcome. First, substantial sunk costs and economies of scale limit entry at the IaaS layer. Second, Microsoft's software licensing practices may create input cost asymmetries that structurally disadvantages rival clouds, consistent with de Cornière and Taylor (2021). Third, switching costs (technical, financial, and organisational) reduce customers ability to switch to alternative providers, even when more competitive offers are available, in line with Klemperer (1995). Together, these mechanisms harm business users, public sector entities exposed to geopolitical risk, and end consumers bearing the downstream pass-through of higher costs.

The analysis suggests that the case for intervention under the DMA is credible, but not uniform across providers. Azure satisfies the qualitative gateway criterion under Article 3(9) more convincingly than AWS. Microsoft's vertically integrated stack, spanning Windows OS, Microsoft 365, and Azure, means business users operate within Microsoft's ecosystem to reach end consumers directly, and prior designations for Windows PC OS and LinkedIn demonstrate the Commission's established willingness to designate qualitatively. This may align more clearly with the DMA's designation of an important gateway. By contrast, AWS's role as primarily infrastructure may make its designation less straightforward, potentially requiring alternative or complementary enforcement approaches. The more appropriate avenue for AWS may therefore be ex post enforcement under Article 102 TFEU, targeting egress fees and switching cost practices that raise dominance concerns without satisfying the gateway criterion.

This asymmetry has direct consequences for remedies. For Azure, qualitative designation unlocks the full Articles 5 and 6 toolkit. For AWS, targeted behavioural intervention under competition law is the more proportionate instrument. Across both providers, as Chen, Doraszelski and Harrington (2009) demonstrate, financial and technical remedies must be implemented simultaneously, reducing egress fees without mandating interoperability risks leaving the tipping dynamic intact. As cloud infrastructure becomes increasingly critical to AI development and Europe's strategic autonomy, the Commission's investigations represent a necessary intervention. However, their effectiveness depends on recognising that AWS and Azure present distinct regulatory challenges that demand differentiated legal treatment.

APPENDIX/GLOSSARY

Cloud Computing Architecture & Services

Cloud Computing: The outsourcing of physical IT infrastructure at scale, where businesses rent computing capacity from third-party providers operating large data centres on demand.

Infrastructure as a Service (IaaS): The foundational layer of cloud computing, referring to the provision of raw computing services, such as servers, storage, and networking, rented by customers over the internet. Examples: AWS Elastic Compute Cloud (EC2), Microsoft Azure Virtual Machines.

Platform as a Service (PaaS): A service operating a layer above IaaS, providing the tools and environment used to develop, test, run, and manage applications in the cloud. Examples: AWS Lambda, Azure App Service.

Software as a Service (SaaS): The provision of finished application software that sits atop the PaaS and IaaS layers. Examples: Microsoft 365, Salesforce.

Hyperscalers: Massive-scale public cloud providers capable of providing the extensive data centers and infrastructure required for modern digital economies and AI development. Examples: Amazon Web Services (AWS), Microsoft Azure, Google Cloud.

On-Premise IT: Traditional IT infrastructure where businesses maintain their own physical servers and storage hardware internally, rather than renting capacity from a cloud provider.

Interoperability Terminology

Application Programming Interface (API): A set of rules and protocols that allows different software applications and cloud services to communicate with each other. Open APIs are critical for cloud interoperability.

Data Egress / Ingress: The technical transfer of data moving out of a cloud provider's network (egress) or into it (ingress). The technical friction and associated egress fees are key for investigations into switching barriers.

Regulatory Terminology

Digital Markets Act (DMA): European Union legislation aimed at ensuring fair, open, and competitive digital markets by imposing regulatory obligations on designated gatekeepers.

Gatekeeper: A designation under Article 3 of the DMA for firms that have a significant impact on the internal market, serve as an important gateway for business users to reach end users, and enjoy an entrenched and durable position.

Core Platform Service (CPS): A distinct category of digital services under the DMA, including cloud computing, that is significant enough to warrant regulation.

FRAND Licensing: An obligation to price and provide access to essential software or intellectual property on "Fair, Reasonable, and Non-Discriminatory" terms.

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